



**RAJIV GANDHI COLLEGE OF ENGINEERING AND
TECHNOLOGY**

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 1

Prepared by: MOHAMED B SIRAJUDEEN

Reg No : 22TD0053

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Unit : I

Teacher's Sign :

Department Vision and Mission.

Vision

Cultivate student as technocrats, researchers and entrepreneurs in the field of computer science and engineering.

Mission

- + To impart quality education in Computer science and engineering by means of learning techniques and value-added courses.
- + To inculcate professional excellence and research culture by encouraging projects in cutting edge technologies through industry interactions.
- + To build leadership skills, ethical values and teamwork among the students.
- + To strengthen the collaboration of department and industry through internships, mentorships and professional body activities.

What is the role of HTTP in web development.

- * HTTP (HyperText Transfer protocol) is the foundation of communication between web browsers (client) and servers, enabling users to access websites and web applications.
- * It follows a request-response model, where the client sends a requested data or an error message.
- * HTTP is the stateless, meaning each request is independent, requiring cookies or tokens to maintain user sessions.

2. Name some commonly used client-side programming language for web development, and write their role in creating interactive user interfaces.

Some commonly used client-side programming languages for web development include.

- * HTML (HyperText Markup Language)

- * CSS (Cascading Style Sheet)

- * JavaScript

- * TypeScript

- * Web Assembly (WASM)

3. State the use of JavaScript in web development.

JavaScript plays a crucial role in web development by adding interactivity, functionality, and dynamic behaviour to websites.

1. Interactive user Interfaces: JavaScript enables features like drop down menus, sliders, modal window and form validators.

2. Client-side form validation: Javascript uses inputs before submission, ensuring data accuracy and reducing server load.

3. Asynchronous communication: IE facilitates data exchange with the server in the background, enabling smooth experience like infinite scrolling and auto-suggestion.

4. Mention the need for CSS.

- * Styling - Enhances colors, fonts and backgrounds.
- + Layout : Organizes content with grids and flex box.
- + Responsive design: Adapts sites to all screen size.
- + Consistency - Maintains a uniform look across pages.
- + Performance : Speeds up load time and simplifies updates.

5) How will you create the password field in the HTML form.

To create a password field in an HTML form, you can use the `<input>` element with the `type = "password"` attribute.

Example: `<form action = "/" submit " method = "post">`
`<label for = "password"> password: </label>`
`<input type = "password" id = "password" name =`

password" required >

<button type = "submit" > submit </button>
</form>

Explanation :

- * The <input> element with type = "password" Create a password field.
- * The id = "password" associates the label with the input field.
- * The name = "password" allows the password value to be sent with the form data.

6) What are literals.

- * Literals in programming are fixed values directly written into the code, representing constant data.
- * They come in various types, including string literals, numeric literals, boolean literals, array literals and object literals.
- * These values are not calculated but used direct in expressions or assignments.
- * Literals provide a simple and efficient way to represent constant data in code.

7) Write a short Note on MIME.

MIME (Multipurpose Internet Mail extensions) is a standard used to extend the format of email message to support text in character sets other

than ASCII, as to support text in character sets other than ASCII, as to support text in character sets other than ASCII, as to support as well as attachments such as images, audio and video.

- + It works by adding headers to email or HTTP messages that specify the type of content being sent

- + MIME types are essential for ensuring that web browsers and email clients can correctly interpret and display different media types.

8) Define Markup Language.

- * A markup language is a type of language used to annotate text or data with additional information, typically to define its structure, presentation or meaning.

- * It uses tags or symbols to identify elements within a document.

- + Examples of markup languages include HTML and XML.

9) Explain the elements of WWW.

The World Wide Web (WWW) is made up of several key elements that work together to deliver content and provide a user-friendly experience. These elements include:

- * Web browsers

- * Web servers

- * Web pages

- * URLs

- * Hyperlinks.

Q) What are the ways by which the server and the browser do communicate,

- 1 The server and browser communicate through HTTP, HTTPS, where the browser sends an HTTP request and the server responds with an HTTP response.
- 1 Cookies and sessions help maintain user state across requests.
- 1 AJAX allow the browser to send background requests and update content without reloading the page.
- 1 Websockets provide real time, bidirectional communication for live updates.

Form in HTML

A form in HTML is a section of a document that allows users to input and submit data.

Forms are used to collect user information, such as login credentials, registration details, feedback, surveys and more.

Forms are created using the `<form>` tag and contain various input elements like text fields, checkboxes, radio buttons, dropdowns, and buttons.

Types of forms in HTML

1. Text input form

Used to collect single-line text input.

```
<input type="text" name="username" placeholder="Enter your name">
```

2. Password input form

Used to collect sensitive information like passwords.

```
<input type="password" name="password" placeholder="Enter your password">
```

3. Radio button form

Used to select one option from multiple choices

```
<input type="radio" name="gender" value="male"> Male  
<input type="radio" name="gender" value="female"> Female
```


4. Checkbox form

Used to select multiple options from a list.

```
<input type = "checkbox" name = "hobby" value = "reading" >
```

```
<input type = "checkbox" name = "hobby" value = "sports" >
```

Reading
Sports.

5. Dropdown / Select form

Used to select one option from a dropdown list.

```
<select name = "country" >
```

```
<option value = "usa" > USA </option>
```

```
<option value = "canada" > Canada </option>
```

```
<option value = "uk" > UK </option>
```

```
</select >
```

6. File upload form

Used to upload files.

```
<input type = "file" name = "fileupload" >
```

7. Submit and Reset form

Used to submit or reset the form data.

```
<input type = "submit" value = "Submit" >
```

```
<input type = "reset" value = "Reset" >
```


Example Program

```
<html lang='en'>
```

```
<head>
```

```
<title> All Types of Forms </title>
```

```
</head>
```

```
<body>
```

```
<h1> All Types of Forms in HTML </h1>
```

```
<form action="#" method="post">
```

```
<!-- Text input -->
```

```
<label for="name">Name: </label>
```

```
<input type="text" id="name" name="name"
```

```
placeholder="Enter your name"><br><br>
```

```
<!-- Password input -->
```

```
<label for="password">Password: </label>
```

```
<input type="password" id="password" name="password"
```

```
placeholder="Enter your password"><br><br>
```

```
<!-- Radio buttons -->
```

```
<label> Gender: </label>
```

```
<input type="radio" id="male" name="gender" value="male">
```

```
<label for="male">Male </label>
```

```
<input type="radio" id="female" name="gender" value="female">
```

```
<input type="radio" id="female" name="gender" value="female">
```

```
<label for="female">Female </label> <br><br>
```

```
<!-- Checkboxes -->
```

```
<label> Hobbies: </label>
```

```
<input type="checkbox" id="reading" name="hobby" value="reading">
```

```
<label for="reading">Reading </label>
```

```
<input type="checkbox" id="travel" name="travel" value="travel">
```

```
<label for="travel">Travel </label>
```


<!-- Dropdown / Select -->

<label for = "country" > Country: </label>

<select id = "country" name = "country">

<option value = ""> Select your country </option>

<option value = "usa"> USA </option>

<option value = "canada"> Canada </option>

<option value = "uk"> UK </option>

</select>

<!-- File upload -->

<label for = "file"> Upload a file: </label>

<input type = "file" id = "file" name = "file">

<!-- Submit and Reset -->

<input type = "submit" value = "Submit">

<input type = "reset" value = "Reset">

</form>

</body>

</html>